

Expressions: Find the Distribution & Like-Term Errors

Answer Key

Grade 6–7 Mathematics

Quick Answers

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|--------------------------|--------------------------|
| 1. $8x$ | 5. (b) = 20, — |
| 2. $6x + 24$ | 6. (b) = 14, — |
| 3. $7x + 5$ | 7. $-6x + 15$ |
| 4. (a) = 21, (b) = 17, — | 8. (a) = 38, (b) = 34, — |
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What is verified

4 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 5, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6–7 Mathematics

7.EE.A – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 14 tagged checks.

Common wrong answers

5. *If they got 11: distributed the 4 to only the first term*
 6. *If they got 2: dropped the variable and kept only 6 - 4*
 8. *If they got 34: distributed the 2 to only the first term*
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1. Combine like terms: $5x + 3x$

Solution: Both terms are x -terms, so add the coefficients and keep the variable: $5+3 = 8$ copies of x . $8x$

2. Expand: $6(x + 4)$

Solution: Distribute the 6 to *both* terms: $6 \cdot x + 6 \cdot 4$.

$6x + 24$

3. Combine like terms: $4x + 7 + 3x - 2$

Solution: x -terms: $4x + 3x = 7x$. Constants: $7 - 2 = 5$.

$7x + 5$

4. Consider $3(x + 2)$ and $3x + 2$. (a) evaluate the first at $x = 5$; (b) evaluate the second at $x = 5$; (c) are they equivalent?

Solution: (a) $3(5 + 2) = 3 \cdot 7 = 21$.

21

(b) $3 \cdot 5 + 2 = 15 + 2 = 17$.

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(c) Model answer: they are *not* equivalent — substituting $x = 5$ gives 21 for one and 17 for the other, and equivalent expressions must give the same value for every x . (Instructor judges the sentence; it must point to the two different values, or to the missing $3 \cdot 2$.)

5. Deshawn evaluates $4(x + 3)$ when $x = 2$ and writes $4 \cdot 2 + 3 = 11$. (a) name the error; (b) find the correct value.

Solution: (a) Deshawn distributed the 4 to only the first term: he multiplied the x by 4 but left the $+3$ alone. The 4 multiplies the *whole* sum $(x + 3)$.

(b) $4(2 + 3) = 4 \cdot 5 = 20$.

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6. Ava simplifies $6x - 4x$ and writes $6x - 4x = 2$ “because the x ’s cancel.” (a) name the error; (b) find the value of $6x - 4x$ when $x = 7$.

Solution: (a) Ava dropped the variable: $6x - 4x$ means 6 copies of x minus 4 copies of x , which leaves $2x$ — two copies of x , not the number 2. The x ’s do not cancel; they are what is being counted.

(b) $6x - 4x = 2x$, and at $x = 7$ that is $2 \cdot 7 = 14$.

14

7. Expand: $-3(2x - 5)$

Solution: Distribute -3 to both terms, watching signs: $-3 \cdot 2x = -6x$ and $-3 \cdot (-5) = +15$.

$-6x + 15$

8. Challenge. Jaylen claims $2(3x + 4) = 6x + 4$. (a) evaluate $2(3x + 4)$ at $x = 5$; (b) evaluate $6x + 4$ at $x = 5$; (c) what did he forget?

Solution: (a) $2(3 \cdot 5 + 4) = 2(15 + 4) = 2 \cdot 19 = 38$.

38

(b) $6 \cdot 5 + 4 = 30 + 4 = 34$.

34

(c) Model answer: Jaylen multiplied only the $3x$ by 2 and left the $+4$ alone; the 4 must be doubled too, so the correct expansion is $6x + 8$. The two different values in (a) and (b) prove his claim wrong. (Instructor judges the sentence.)